

Due: Thursday February 25th, latest at 2PM

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This problem set is about gradients, divergences and curls in electrostatics, i.e., about the relationships between electric field, electric potential and electric charge density.

You should know Poisson's equation ($\nabla^2 V = -\rho/\epsilon_0$), Maxwell's first equation ($\nabla \cdot \mathbf{E} = \rho/\epsilon_0$), and of course that the electric field is the negative gradient of the electric potential.

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1. **[6 points]** Using cartesian coordinates, show that the curl of the gradient of a scalar field is zero, i.e., $\nabla \times (\nabla f) = 0$ for any scalar field $f(x, y, z)$.

[Note: this means: a vector field that can be expressed as a gradient of a scalar field, must have zero curl. Thus the electric field, which is the gradient of $-V$, must have zero curl: $\nabla \times \mathbf{E} = 0$.]

2. The electric potential in some region of space is given by

$$V(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

or, in spherical coordinates, $V(r) = 1/r$.

- (a) **[6 points]** Calculate an expression for the electric field in this region, using cartesian coordinates.
 - (b) **[5 points]** Noting that $x\hat{i} + y\hat{j} + z\hat{k} = \mathbf{r} = r\hat{r}$, express the expression for $\mathbf{E}$ in spherical coordinates.
 - (c) **[5 points]** Look up the expression for the gradient operator in spherical coordinates. Re-derive the expression for $\mathbf{E}$, now directly in spherical coordinates.
 - (d) **[5 points]** Look up the expression for the Laplacian operator in spherical coordinates. By calculating the Laplacian of V , find out the charge density ρ in this region.
 - (e) **[5 points]** Calculate the Laplacian of V (and hence the charge density ρ) using cartesian coordinates. (This is a cumbersome calculation.)
3. **[6 points]** Show that the vector field $\mathbf{A} = y\hat{i} - x\hat{j}$ cannot be a valid electric field, because it's curl is nonzero. Which direction does $\nabla \times \mathbf{A}$ point in?

4. The electric field is claimed to be $\mathbf{E} = x\hat{i} + y\hat{j}$ in some region of space.
- (a) **[6 points]** Show that this satisfies the requirement of zero curl.
 - (b) **[6 points]** Using Maxwell's first equation, find the electric charge density ρ in this region of space. How does ρ vary in space?